

CBSE 2027 Physics — Predicted Paper (Set 2 of 10)

SECTION A: Objective Type Questions (16 Marks)

This section consists of 12 MCQs and 4 Assertion-Reasoning questions of 1 mark each.

Q1. What is the electric flux through a cube of side **1 cm** which encloses an electric dipole? (a) q/ϵ_0 (b) Zero (c) $2q/\epsilon_0$ (d) $q/6\epsilon_0$ (Chapter: Electric Charges and Fields) | **[Confidence: Very High]** [1]

Q2. By increasing the temperature, the specific resistance of a conductor and a semiconductor respectively: (a) increases for both. (b) decreases for both. (c) increases for a conductor and decreases for a semiconductor. (d) decreases for a conductor and increases for a semiconductor. (Chapter: Current Electricity) | **[Confidence: Very High]** [2]

Q3. Two wires of the same length are shaped into a square of side ' a ' and a circle with radius ' r '. If they carry the same current, the ratio of their magnetic moments is: (a) $2 : \pi$ (b) $\pi : 2$ (c) $\pi : 4$ (d) $4 : \pi$ (Chapter: Moving Charges and Magnetism) | **[Confidence: High]** [3]

Q4. The susceptibility of a magnetic material is -4.2×10^{-6} . The material is: (a) ferromagnetic (b) paramagnetic (c) diamagnetic (d) none of the above (Chapter: Magnetism and Matter) | **[Confidence: Very High]** [4]

Q5. Displacement current exists only when: (a) electric field is changing. (b) magnetic field is changing. (c) electric field is not changing. (d) magnetic field is not changing. (Chapter: Electromagnetic Waves) | **[Confidence: Very High]** [5]

Q6. Which of the following graphs represents the correct variation of capacitive reactance X_C with frequency ν ? (a) A straight line passing through the origin. (b) A rectangular hyperbola (inversely proportional curve). (c) A parabola. (d) A straight line parallel to the frequency axis. (Chapter: Alternating Current) | **[Confidence: High]** [6]

Q7. A biconcave lens of power P vertically splits into two identical plano-concave parts. The power of each part will be: (a) $2P$ (b) $P/2$ (c) P (d) $P/\sqrt{2}$ (Chapter: Ray Optics and Optical Instruments) | **[Confidence: Medium]** [7]

Q8. In a single slit diffraction experiment, if the width of the slit is increased, the size of the central bright band will: (a) increase (b) decrease (c) remain unchanged (d) first increase then decrease (Chapter: Wave Optics) | **[Confidence: Very High]** [8]

Q9. The kinetic energy of a proton and that of an α -particle are **4 eV** and **1 eV**, respectively. The ratio of de-Broglie wavelengths associated with them will be: (a) **2 : 1** (b) **1 : 1** (c) **1 : 2** (d) **4 : 1** (Chapter: Dual Nature of Radiation & Matter) | **[Confidence: High]** [9]

Q10. When two nuclei ($A \leq 10$) fuse together to form a heavier nucleus, the: (a) binding energy per nucleon increases. (b) binding energy per nucleon decreases. (c) binding energy per nucleon does not change. (d) total binding energy decreases. (Chapter: Nuclei) | **[Confidence: Very High]** [10]

Q11. In an n-type semiconductor, the donor energy level lies: (a) at the centre of the energy gap. (b) just below the conduction band. (c) just above the valence band. (d) in the conduction band. (*Chapter: Semiconductor Electronics*) | **[Confidence: Very High]** [11]

Q12. The power factor of an A.C. circuit is 0.5. What is the phase difference between voltage and current in the circuit? (a) $\pi/2$ (b) $\pi/4$ (c) $\pi/3$ (d) $\pi/6$ (*Chapter: Alternating Current*) | **[Confidence: High]** [12]

Q13. Assertion (A): A negative charge in an electric field moves along the direction of the electric field. **Reason (R):** On a negative charge, a force acts in the direction of the electric field. (*Chapter: Electric Charges and Fields*) | **[Confidence: Very High]** [13]

Q14. Assertion (A): Kirchhoff's junction rule is based on the conservation of charge. **Reason (R):** A resistor obeys Ohm's law while a diode does not. (*Chapter: Current Electricity*) | **[Confidence: High]** [14]

Q15. Assertion (A): When the radius of a circular wire carrying current is doubled, its magnetic moment becomes four times. **Reason (R):** The magnetic moment of a current-carrying loop is directly proportional to the area of the loop. (*Chapter: Moving Charges and Magnetism*) | **[Confidence: Very High]** [15]

Q16. Assertion (A): For radiation of a frequency greater than the threshold frequency, the photoelectric current is proportional to the intensity of the radiation. **Reason (R):** Greater the number of energy quanta available, greater is the number of electrons absorbing the energy quanta and greater is the number of electrons coming out of the metal. (*Chapter: Dual Nature of Radiation & Matter*) | **[Confidence: High]** [16]

SECTION B: Very Short Answer Type Questions (10 Marks)

This section consists of 5 questions of 2 marks each.

Q17. Two identical conducting balls **A** and **B** have charges $-Q$ and $+3Q$ respectively. They are brought in contact with each other and then separated by a distance d apart. Find the nature and magnitude of the Coulomb force between them. (*Chapter: Electric Charges and Fields*) | **[Confidence: Very High]** [1]

Q18. Relative permeability of a material is $\mu_r = 0.5$. Identify the nature of the magnetic material and write its relation to magnetic susceptibility. (*Chapter: Magnetism and Matter*) | **[Confidence: High]** [4]

Q19. A metallic rod **PQ** of length l is rotated with an angular velocity ω about an axis passing through its mid-point (**O**) and perpendicular to the plane of the paper, in a uniform magnetic field **B**. What is the potential difference developed between the two ends of the rod, **P** and **Q**? (*Chapter: Electromagnetic Induction*) | **[Confidence: High]** [17]

Q20. When an object is placed between f and $2f$ of a concave mirror, would the image formed be (i) real or virtual, and (ii) diminished or magnified? (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: Very High]** [7]

Q21. If the frequency of light incident on the cathode of a photocell is increased, how will the following be affected? Justify your answer. (A) Kinetic Energy of photoelectrons. (B) Photocurrent. (*Chapter: Dual Nature of Radiation and Matter*) | **[Confidence: Very High]** [18]

SECTION C: Short Answer Type Questions (21 Marks)

This section consists of 7 questions of 3 marks each.

Q22. Define the term 'mobility' of charge carriers in a current-carrying conductor. Obtain the relation for mobility in terms of relaxation time. (*Chapter: Current Electricity*) | **[Confidence: Very High]** [2]

Q23. Two identical circular coils, P and Q each of radius R , carrying currents 1 A and $\sqrt{3}\text{ A}$ respectively, are placed concentrically and perpendicular to each other lying in the XY and YZ planes. Find the magnitude and direction of the net magnetic field at the centre of the coils. (*Chapter: Moving Charges and Magnetism*) | **[Confidence: High]** [19]

Q24. An AC voltage $V = V_0 \sin(\omega t)$ is applied across a pure inductor of inductance L . Find an expression for the current flowing in the circuit and show mathematically that the current flowing through it lags behind the applied voltage by a phase angle of $\pi/2$. (*Chapter: Alternating Current*) | **[Confidence: Very High]** [12]

Q25. Describe how displacement current is produced between the plates of a parallel plate capacitor during charging. Depict the field's diagram of an electromagnetic wave propagating along the positive X-axis with its electric field along the Y-axis. (*Chapter: Electromagnetic Waves*) | **[Confidence: High]** [20]

Q26. A beam of light consisting of two wavelengths 600 nm and 500 nm is used in a Young's double slit experiment. The slit separation is 1.0 mm and the screen is kept 0.60 m away from the plane of the slits. Calculate: (A) The distance of the second bright fringe from the central maximum for the wavelength 500 nm . (B) The least distance from the central maximum where the bright fringes due to both the wavelengths coincide. (*Chapter: Wave Optics*) | **[Confidence: Very High]** [8, 21]

Q27. The total energy of an electron in the first excited state of the hydrogen atom is about -3.4 eV . (A) What is the kinetic energy of the electron in this state? (B) What is the potential energy of the electron in this state? (C) Which of the answers above would change if the choice of the zero of potential energy is changed? (*Chapter: Atoms*) | **[Confidence: Very High]** [22]

Q28. Write the two processes that take place in the formation of a $p - n$ junction. Explain with the help of a diagram, the formation of the depletion region and barrier potential in a $p - n$ junction. (*Chapter: Semiconductor Electronics*) | **[Confidence: Very High]** [23]

SECTION D: Case-Study Based Questions (8 Marks)

This section consists of 2 questions of 4 marks each. Questions usually have internal sub-parts.

Q29. Case Study 1: V-I Characteristics of a Diode *Passage:* In forward bias measurement, we use a milliammeter since the expected current is large, while a microammeter is used in reverse bias to measure the current. In forward bias, the current first increases very slowly, almost negligibly, till the voltage across the diode crosses a certain value. After this characteristic voltage, the diode current increases significantly (exponentially). This voltage is called the threshold voltage or cut-in voltage (0.2 V for germanium diode and ~0.7 V for silicon diode). (a) In a $p - n$ junction, the potential barrier is due to the charges on either side of the junction. What are these charges? (1 mark) (b) How does the resistance of the semiconductor diode behave in forward bias? (1 mark) (c) In an unbiased $p - n$ junction, why is the net junction current at equilibrium zero? **OR** What is the term for the voltage at which forward bias current increases rapidly? Explain its physical significance. (2 marks) (*Chapter: Semiconductor Electronics*) | **[Confidence: High]** [11, 24]

Q30. Case Study 2: Optical Instruments *Passage:* A number of optical devices and instruments have been designed and developed such as periscopes, binoculars, microscopes, and telescopes utilising the reflecting and refracting properties of mirrors, lenses, and prisms. Our knowledge about the formation of images by mirrors and lenses is the basic requirement for understanding the working of these devices. (a) Why is the image formed at infinity often considered most suitable for viewing through optical instruments? (1 mark) (b) In modern microscopes, multi-component lenses are used for both the objective and the eyepiece. Why? (1 mark) (c) Write two distinct advantages of a reflecting-type telescope over a refracting-type telescope. **OR** Write two points of difference between a compound microscope and an astronomical telescope. (2 marks) (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: Very High]** [25]

SECTION E: Long Answer Type Questions (15 Marks)

This section consists of 3 questions of 5 marks each. All questions have an internal "OR" choice.

Q31. (a) State Gauss's Law in electrostatics. Use it to obtain an expression for the electric field due to an infinitely long thin straight wire with uniform linear charge density λ . **(b)** An infinitely long positively charged straight wire has a linear charge density λ . An electron is revolving in a circle with a constant speed v such that the wire passes through the centre, and is perpendicular to the plane of the circle. Find the kinetic energy of the electron in terms of the magnitudes of its charge and the linear charge density λ . **OR (a)** Derive an expression for the electric field at any point on the equatorial line of an electric dipole. **(b)** Two identical point charges, q each, are kept 2 m apart in air. A third point charge Q of unknown magnitude and sign is placed on the line joining the charges such that the system remains in equilibrium. Find the position and nature of Q . (*Chapter: Electrostatics / Electric Charges and Fields*) | **[Confidence: Very High]** [26, 27]

Q32. (a) State the underlying principle of a moving coil galvanometer. **(b)** Explain, using a labelled diagram, how a radial magnetic field is achieved and explain its significance when a

current-carrying coil is kept in it. **(c)** Define the terms: (i) voltage sensitivity and (ii) current sensitivity of a galvanometer. **OR (a)** Draw a labelled diagram of an AC generator. **(b)** Obtain the expression for the instantaneous value of the emf induced in the rotating coil of N turns, each of cross-sectional area A , in the presence of a uniform magnetic field B . **(c)** A horizontal conducting rod 10 m long extending from east to west is falling with a speed 5.0 ms^{-1} at right angles to the horizontal component of the Earth's magnetic field, $0.3 \times 10^{-4}\text{ Wb m}^{-2}$. Find the instantaneous value of the emf induced in the rod. (*Chapter: Moving Charges and Magnetism / Electromagnetic Induction*) | **[Confidence: Very High]** [28, 29]

Q33. (a) Draw a ray diagram for the formation of the image of a point object by a thin double convex lens having radii of curvature R_1 and R_2 . Hence, derive the lens maker's formula. **(b)** A converging lens has a focal length of 10 cm in air. It is made of a material of refractive index 1.6 . If it is immersed in a liquid of refractive index 1.3 , find its new focal length. **OR (a)** State Huygens' principle. With the help of a suitable diagram, show how a plane wavefront is refracted from a denser to a rarer medium. Hence, verify Snell's law of refraction. **(b)** A ray of light travels from a medium into water at an angle of incidence of 18° . The refractive index of the medium is more than that of water and the critical angle for the interface between the two media is 20° . Under what conditions does the phenomenon of total internal reflection take place? (*Chapter: Ray Optics / Wave Optics*) | **[Confidence: Very High]** [30, 31]

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